

G H Raisoni College of Engineering & Management, Pune <i>(An Autonomous Institute affiliated to SP Pune University)</i>
S.Y. B. Tech. /Semester-III /Div-A/2023-24
Department of Data Science
UDSL202: Database Management Systems

Question Bank

Unit 3

What is Functional Dependency? Explain types and properties of FD's.

Explain the concept of functional dependency with an example.

What is functional dependency? Explain the inference rules for functional dependency with proof.

What is Functional Dependency? Explain types and properties of FD's.

What is normalization? Explain its need.

Discuss in detail about various normal forms.

Why normalization is needed? Explain the process of normalization.

What is meant by the closure of functional dependencies? Illustrate with an example.

Mention the desirable properties of relation decomposition.

Describe lossless join decomposition.

Show how to preserve Functional Dependencies during decomposition.

What do you understand by attribute closure? Give an example.

Explain insertion, deletion, and modification anomalies.

What is an objective of the normalization?

What is need of normalization? Explain 1NF , 2NF, 3 NF with examples.

Explain 3NF and BCNF. What is the difference between them.

By means of an example show how BCNF is stronger than 3NF

Define multivalued dependency? Explain 4NF with example.

Define 4NF. When it is violated? Why is it useful

Define JOIN dependency. Explain 5NF with an example.

Unit 4

Illustrate transaction properties.

What is transaction? Mention the desirable properties of a transaction.

How to perform rollback, commit, check pointing operations on transactions? Explain

Explain ACID properties of database transaction

State and explain two-phase commit protocol.

What is serialization? Explain it.

Explain about deadlocks.

Compare wait/die with wound/wait scheme

Describe Wait/Die & Wound/Wait protocols.

Describe Wait/Die and Wound/Wait deadlock protocols

Write about the transaction management with SQL using commit, rollback, and savepoint.

How to perform rollback, commit, check pointing operations on transactions? Explain.

Explain concurrency control with locking methods.

Why do we need locks? Explain

Explain two phase locking technique for concurrency control.

What is schedule ?illustrate with example.

What is need of lock in DBMS? Explain shared lock and exclusive lock with the help of example

Show how 2PL protocol ensures serializability.

Explain view Serializability with an example? How it is different from conflict Serializability?

Explain read-only, write-only and read-before-write protocols in serializability.

Explain Two Phase- Locking protocol .What benefit does strict two-phase locking protocol provides?

Discuss its disadvantages.

When deadlock problem occurred? Explain how this problem can be resolved?

Describe the concept of cursor and how it is used in SQL

Explain stored procedures in sql with example.

Explain the term stored procedures and give examples why stored procedures are useful?

How are triggers defined in sql Explain with example.

What is a trigger? How to create it? Discuss various types of triggers

What is a database trigger? Give an example of trigger definition.

Explain Two Phase- Locking protocol .What benefit does strict two-phase locking protocol provides?

Discuss its disadvantages.